

Task1:

Defects and inconsistencies inspection forms

Defect #	Location	Defect type	Classification	Description	Status	Date corrected
1	Stakeholder Description/Stakeholders Summary (Page2)	Omission	Major	Missing key Stakeholders to illustrate the requirements		
2	Product Overview/Product Perspective (Page3)	unintelligibility	Major	Diagrams explaining Backend server and other key aspects are not represented correctly.		
3	Positioning/Product Position Statement (Page2)	ambiguity	Minor	Not sure if E-Concordia Drive is a web-based application or mobile app or both.		
4	Introduction/Scope (Page1)	inadequacy	Major	Scope of the project is not clear if it's dealing with beginner/intermediate/expert level of courses.		
5	Product Feature (Page 4)	poor structuring	Minor	Course not bifurcated as per user competency level.		

6	Other Product Requirements (Page 6)	poor modifiability	Major	Tool is dependent on the database version. If the database needs to be updated then PHP version needs to be updated accordingly.		
7	Other Product Requirements (Page 6)	Omission	Major	Glossary of Terms is not added for better understanding the term.		

#	Location	Inconsistency type	Classification	Description	Status	Date corrected
1	Product Feature (Page 4) S1	Designation Clash	Weak	Admin is editing/updating/approving/publishing the content on E-Concordia Drive.		
2	Product Feature (Page4) S2	Designation Clash	Weak	Trainer also adds/updating/editing the content on E-Concordia Drive.		
3	Stakeholder Descriptions/ Stakeholder Summary (Page2) S3	Designation Clash	Weak	Professor/Lecturer E-Concordia Drive's content approver and publisher		
4	Product Feature (Page 4) S4	Structure Clash	Weak	Login Page for Admin prompts for username and Password		

5	Product Feature (Page 4) S5	Structure Clash	Weak	The Login Page for Trainer also prompts for username and Password which is the same as Admin. It Doesn't provide bifurcation between roles and responsibilities.		
6	Assumptions and Dependencies (Page 3) S6	Structure Clash	Weak	User requires access to a laptop/desktop to take the lessons		
7	Assumptions and Dependencies (Page 3) S7	Structure Clash	Weak	User should be a student of Concordia University to take lessons.		
8	Product Overview/ Product Perspective S8	Structure Clash	Weak	The block diagram depicts how the system works as a whole, highlighting the high-level features.		
9	Other Product Requirements/ Platform Requirement S9	Structure Clash	Weak	PHP MVC based project with MYSQL database structure		
10	Introduction/Scope (Page1) S10	Terminology Clash	Strong	E-Concordia Drive is an E-learning platform for students		

Task2:

S1: Admin is editing/updating/approving/publishing the content on E-Concordia Drive.

S2: Trainer also adding/updating/editing the content on E-Concordia Drive.

S3: Professor/Lecturer E-Concordia Drive's content approver and publisher

S4: Login Page for Admin prompts for username and Password

S5: Login Page for Trainer also prompts for username and Password

S6: User requires access to a laptop/desktop to take the lessons.

S7: User should be a student of Concordia University to take lessons.

S8: The block diagram depicts how the system works as a whole, highlighting the high-level features.

S9: PHP MVC based project with MYSQL database structure.

S10: E-Concordia Drive is an E-learning platform for students

Statement	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Total
S1	0	1	1	0	0	0	0	1000	0	1000	2002
S2	1	0	1	0	0	0	0	1000	0	1000	2002
S3	1	1	0	0	0	0	0	0	0	0	2
S4	0	0	0	0	1	0	0	0	0	1000	1001
S5	0	0	0	1	0	0	0	0	0	1000	1001
S6	0	0	0	0	0	0	1	0	0	1000	1001
S7	0	0	0	0	0	1	0	0	0	1000	1001
S8	1000	1000	0	0	0	0	0	0	1000	1000	4000
S9	0	0	0	0	0	0	0	1000	0	0	1000
S10	1000	1000	0	1000	1000	1000	1000	1000	0	0	7000
Total	2002	2002	2	1001	1001	1001	1001	4000	1000	7000	20100

Task3: Conflict Resolution

1. Consider the divergence between the statements (S1, S2) Admin is editing/updating/approving/publishing the content on E-Concordia Drive' and Trainer also updating/editing the content on E-Concordia Drive'.

Avoiding Boundary Condition	The boundary condition for strong conflict was seen to be 'Both admin and trainer updating the same content'. Avoiding this boundary condition might be achieved by keeping the content session locked for a time period when someone editing it and another person got the notification if they tried to edit.
Restore Conflicting Statements	This conflict can be resolved by restoring conflicting statements by allowing trainers to add/update the contents only after Admin's notification.

2. Consider the divergence between the statements (S1, S3) Admin is editing/updating/approving/publishing the content on E-Concordia Drive' and 'Professor/Lecturer E-Concordia Drive's content approver and publisher'

Avoiding Boundary Condition	The boundary condition for strong conflict was seen to be 'Both Admin and Professor approving/publishing the same content at the same time'. Avoiding this boundary condition might be achieved by restricting their access. Professor is allowed only to provide updated content to Trainer and Admin should have the access for final approval and publishing.
Weaking Conflicting Statements	Weaken Conflicting statements by allowing the Admin to delegate the content approver or publishing access to the professor. This will ensure the Subject Matter Expert to review and publish the content. Also reduces the workload of Admin.

3. Consider the two statements 'Login Page for Admin prompts for username and Password' and 'Login Page for Trainer also prompts for username and Password'.

(S4, S5) Statements are conflicting as both are prompting for username and password. There is no User Type option to differentiate the Trainer and Admin Login Page.

Avoiding Boundary Condition	Avoid Boundary Condition by providing a separate login portal for Trainer and Admin which will be verifying the credentials from separate database tables.
Weakening Conflicting Statements	Weaken Conflicting statements by introducing the new requirement of adding usertype option to differentiate the different user's Login Page. Assigning unique UserID to every user and maintaining their user type in the database. Based on which they will be able to perform their respective activities. Resolved Statement would be 'Login Page prompts for username, Password and User Type'.

4. Consider the two statements 'User requires access to a laptop/desktop to take the lessons' and 'User should be a student of Concordia University to take lessons.'

(S6, S7) Statements are conflicting as one statement states that only access to a laptop/desktop is required to take the lessons and another defines Student ID required to access the lessons.

Specializing Conflict	This conflict can be resolved by specializing the conflict source object "access through laptop/desktop" into "access through laptop/desktop using Student ID". Resolved Statement: 'Users should have the access to the laptop/desktop and Student Id to access the driving lessons and quizzes.'
Weaken Conflicting Statement	This conflict can be weakened by providing a direct access link of E-Concordia Drive from Concordia's Student portal. Easy to handle and manage (multiple links/passwords) for students.

5. Consider the two statements 'The block diagram depicts how the system works as a whole, highlighting the high-level features' and 'PHP MVC based project with MYSQL database structure'.

(S8, S9) Statements are conflicting as one statement states that the block diagram depicts how the system works without including the environment and server details and other statements clearly define the project as a PHP MVC model with MYSQL database structure.

Specializing the Conflict	This conflict can be resolved by Specializing the Conflict Source by highlighting the block diagram which depicts how the system works using PHP MVC Model with MYSQL database structure. Resolved Statement: 'The block diagram depicts how the system works as a whole, highlighting the high-level features using the PHP MVC model with MYSQL database structure'
Drop Lower-Priority Statements	Drop lower-priority statements by reducing the priority of specifications of Project tools and technology as initially we are more focused on highlighting the overall functionality of end users.

6. Consider the statement 'E-Concordia Drive is an E-learning platform for students'

The S10 statement is very conflicting in the sense that it doesn't give a clear idea if E-Concordia Drive is a web application or mobile application or supports both.

Specializing Conflict	This conflict can be resolved by specializing in the conflict source object of E-Concordia Drive. Resolved Statement: 'E-Concordia Drive is an E-learning web application platform for students'
Weaking Conflicting Statements	Weaken Conflicting Statement by adding the specification that the portal is compatible to both laptop/desktop and mobile based browsers.

Task4: Conflict Evaluation

Using Weighted matrices for evaluating alternative options for the above documented conflicts.

$$totalScore(opt) = \sum_{crit} (Scores(opt, crit) \times Weight(crit))$$

1. Consider the conflicts between the statements (S1, S2) Admin is editing/updating/approving/publishing the content on E-Concordia Drive' and 'Trainer also updating/editing the content on E-Concordia Drive'

		Option Scores	
Evaluation Criteria NFR	Significance Weighting	Content Session locked when admin/trainer updating	Trainer add/update tutorials only after admin's notification
Data Availability	0.6	0.7	0.9
Fast Response	0.3	0.8	0.7
Less Cost	0.1	0.5	0.6
Total	1.0	0.71	0.81

The option “Trainer add/update tutorials only after admin’s notification” seems to be a better option according to above estimates.

2. Consider the conflicts between the statements (S1, S3) Admin is editing/updating/approving/publishing the content on E-Concordia Drive' and 'Professor/Lecturer E-Concordia Drive’s content approver and publisher’

		Option Scores	
Evaluation Criteria NFR	Significance Weighting	Restricting Admin/Professor's role, responsibilities and access	Admin delegate access to Professor if required
Fast response	0.2	0.7	0.8
Reliable	0.2	0.6	0.6
Data Integrity	0.6	0.9	0.7
Total	1.0	0.80	0.70

The option “Restricting Admin/Professor’s role, responsibilities and access” seems to be a better option according to above estimates.

3. Consider the conflicts between two statements ‘Login Page for Admin prompts for username and Password’ and ‘Login Page for Trainer also prompts for username and Password’.

		Option Scores	
Evaluation Criteria NFR	Significance Weighting	Separate login portal for Trainer and Admin with different Database	Adding User Type option to differentiate Login Pages
Usability	0.5	0.7	0.8
Minimal inconvenience	0.2	0.6	0.7
Confidentiality	0.3	0.9	0.7
Total	1.0	0.74	0.75

The option “Adding User Type option to differentiate Login Pages” seems to be a better option according to above estimates.

4. Consider the conflicts between two statements ‘User requires access to a laptop/desktop to take the lessons’ and ‘User should be a student of Concordia University to take lessons.’

		Option Scores	
Evaluation Criteria NFR	Significance Weighting	Access through laptop/desktop using Student ID	Access through Concordia Student Portal
Less Maintainability	0.6	0.7	0.9
Fast Response	0.3	0.9	0.6
Minimal inconvenience	0.1	0.7	0.7
Total	1.0	0.76	0.79

The option “Access through Concordia Student Portal” seems to be a better option according to above estimates.

- Consider the conflicts between two statements ‘The block diagram depicts how the system works as a whole, highlighting the high-level features without including server and model details’ and ‘PHP MVC based project with MYSQL database structure’.

		Option Scores	
Evaluation Criteria NFR	Significance Weighting	Block Diagram highlighting server details, models and key aspects	Block Diagram only highlighting key features
Distribution	0.4	0.8	0.6
Less Maintainability	0.2	0.6	0.7
Software Interoperability	0.4	0.9	0.6
Total	1.0	0.80	0.62

The option “Block Diagram highlighting server details, models and key aspects” seems to be a better option according to above estimates.

- Consider the conflicting statement ‘E-Concordia Drive is an E-learning platform for students’

		Option Scores	
Evaluation Criteria NFR	Significance Weighting	E-Concordia supports Web Platform	E-Concordia supports Web Platform as well as mobile app
Less Cost	0.3	0.8	0.6
Less Maintainability	0.3	0.9	0.7
Minimal Inconvenience	0.4	0.6	0.7
Total	1.0	0.75	0.67

The option “E-Concordia supports Web Platform” seems to be a better option according to above estimates.

Task5: Risk Management

1. Risk Identification

a) Component Inspection:

MYSQL Database server Failure

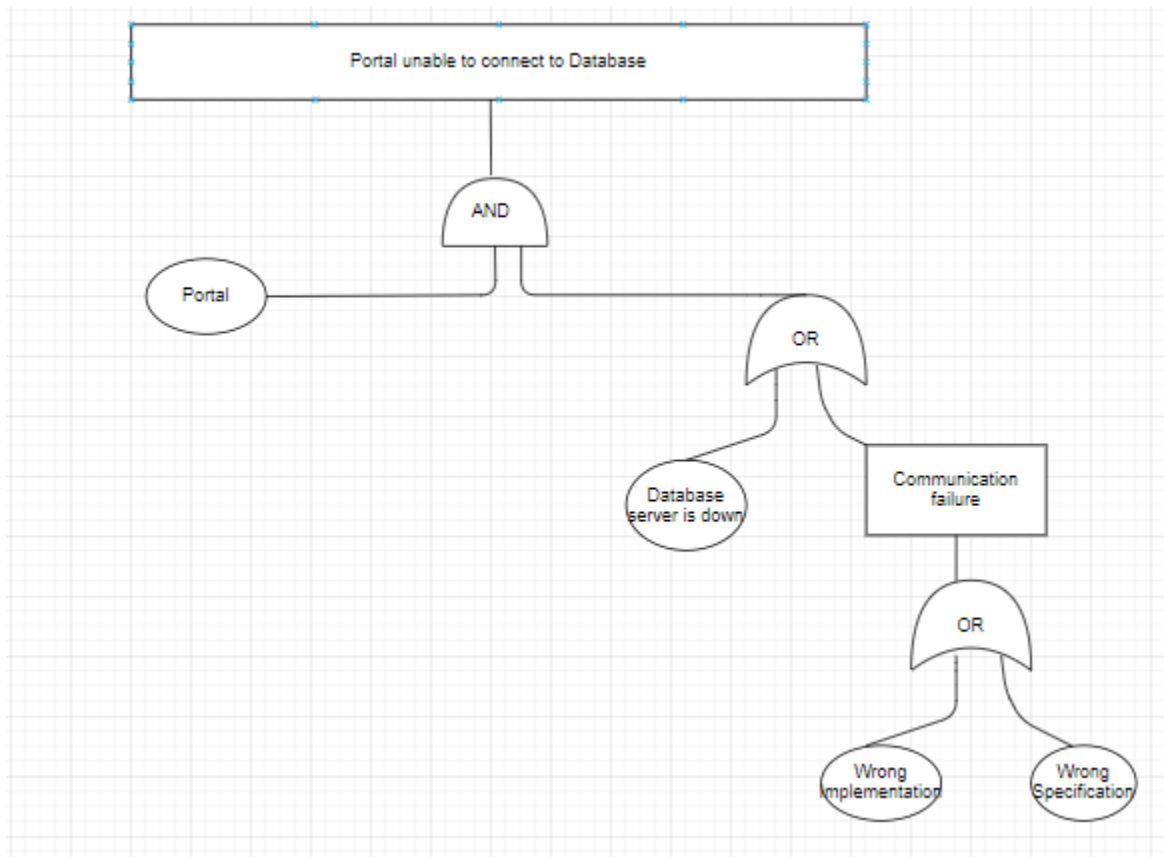
Currently we are using a single MYSQL data structure as a single node which is a single point of failure.

b) Risk Checklist

- 1) Reliability Risk : Currently we are relying solely on Trainer’s Driving knowledge to create/update the driving lessons and quizzes which is not at all reliable to trust.
- 2) Security Risk : Currently E-Concordia Drive is not using any encryption techniques to protect their data. Therefore, it has very high chances of data breaches or hacking.
- 3) Performance Risk : Currently the system has low performance of sending updated pages to a student within five minutes of new content being published to the site. This response time needs to be less than 30 secs.

c) Risk Trees

Communication Risk: Portal unable to connect to MYSQL database.



2. Risk Quantitative Assessment

Risk	Rationale	Likelihood	Severity (1-10)	Risk Exposure (Likelihood*Severity)
Server Failure Risk	MYSQL Database server is a single point of failure. There are very high chances of server failure in case of heavy load (LIKELY)	30%	8	2.4
Reliability Risk	Relying solely on Trainer's Driving knowledge Due to which there is possibility of not providing updated or good material to the Students (POSSIBLE)	20%	6	1.2
Security Risk	There are very high chances of data breaches and hacking if not using any encryption techniques (LIKELY)	35%	8	2.8

Performance Risk	There is a possibility that updated page reflects to a student within five minutes of new content being published to the site (POSSIBLE)	24%	7	1.68
Communication Risk	There are high chances of communication failure between Portal and Database server (LIKELY)	40%	8	3.2

3. Risk Control

We will be using Risk Reduction Leverage (RRL) to calculate the better countermeasure for a particular risk.

$$\text{Risk reduction leverage (RRL)} = \frac{RE_{\text{before}} - RE_{\text{after}}}{\text{Cost of risk reduction}}$$

$$RE = \text{risk probability} \times \text{amount at stake}$$

a) Server Failure Risk

Risk Reduction Tactic: **Reduce Risk Likelihood** by opting below 2 countermeasure alternatives.

Problem Statement: Currently we are using a single MYSQL database as a single node which is a single point of failure in case of heavy load or due to any other issue. The probability of data loss is 30%. The cost of losing data will be measured in terms of cost of its reproduction which is estimated as 8.

Countermeasure alternative1 and 2:

1)Add a MySQL 3 node database cluster to handle increased traffic and provide replication. So that other server will always be running in case of failure. This option will reduce the risk to 4% and is estimated to cost 3.

2)Monitoring tools such as Dynatrace could be used to monitor the data and accordingly notification should be sent to the concerned team at the right time (when a particular threshold crossed). This option will reduce the risk to 5% and is estimated to cost 2.

$$RRL1 = (0.30 \times 8 - 0.04 \times 8) / 3 = (2.4 - 0.32) / 3 = 0.69$$

$$RRL2 = (0.30 \times 8 - 0.05 \times 8) / 2 = (2.4 - 0.4) / 2 = 1$$

Since $RRL2 > RRL1$ which indicates cost effective risk reduction measure. Second alternative is the better option.

b) Reliability Risk

Risk Reduction Tactic: **Reduce Consequence Likelihood** by opting below 2 counter measure alternatives.

Problem Statement: Currently we are relying solely on Trainer's Driving knowledge to create/update the driving lessons and quizzes which is not at all reliable to trust. The probability of not having updated tutorials is 20%. The cost of providing updated materials will be measured in terms of content creation which is estimated as 6.

Countermeasure alternative1 and 2:

1)Tie up with Government Authorities to stay updated with the latest Driving laws and regulation. This option will reduce the risk to 1% and is estimated to cost 1.

2)New Department should be set up to analyze, normalize and integrate the data. This option will reduce the risk to 2% and is estimated to cost 2.

$$RRL1 = (0.20*6 - 0.01*6)/1 = (1.2 - 0.06) / 1 = 1.14$$

$$RRL2 = (0.20*6 - 0.02*6)/2 = (1.2 - 0.12) / 2 = 0.54$$

Since $RRL1 > RRL2$ which indicates cost effective risk reduction measure. First alternative is the better option.

c) Security Risk

Risk Reduction Tactic: **Reduce Risk Likelihood** by opting below 2 counter measure alternatives.

Problem Statement: Currently E-Concordia Drive is not using any encryption techniques to protect their data. Therefore, it has very high chances of data breaches or hacking. The probability of data breaches is 35%. The cost of losing data will be measured in terms of cost of its reproduction which is estimated as 8.

Countermeasure alternative1 and 2:

1)Add software to Run vulnerability scans every fortnight. This option will reduce the risk to 3% and is estimated to cost 2.

2) Add some hash tables, salt algorithms or cryptography to protect your critical data. This option will reduce the risk to 2% and is estimated to cost 3.

$$RRL1 = (0.35 \times 8 - 0.03 \times 8) / 2 = (2.8 - 0.24) / 2 = 1.28$$

$$RRL2 = (0.35 \times 8 - 0.02 \times 8) / 3 = (2.8 - 0.16) / 3 = 0.88$$

Since $RRL1 > RRL2$ which indicates cost effective risk reduction measure. First alternative is the better option.

d) Performance Risk

Risk Reduction Tactic: **Reduce Risk Likelihood** by opting below 2 counter measure alternatives.

Problem Statement: Currently the system has low performance of sending an updated page to a student within five minutes of new content being published to the site. This response time needs to be less than 30 secs. The probability of viewing old data is 24%. The cost of viewing updated data will be measured in terms of the cost of its refreshing which is estimated as 7.

Countermeasure alternative1 and 2:

1) Distribute Traffic with a Load Balancer. In this way data will be transmitted fast to the data server. This option will reduce the risk to 1% and is estimated to cost 1.

2) Deliver Content Faster by caching. This option will reduce the risk to 3% and is estimated to cost 2.

$$RRL1 = (0.24 \times 7 - 0.01 \times 7) / 1 = (1.68 - 0.07) / 1 = 1.31$$

$$RRL2 = (0.24 \times 7 - 0.03 \times 7) / 2 = (1.68 - 0.21) / 2 = 0.73$$

Since $RRL1 > RRL2$ which indicates cost effective risk reduction measure. First alternative is the better option.

e) Communication Risk

Risk Reduction Tactic: **Reduce Risk Likelihood** by opting below 2 counter measure alternatives.

Problem Statement: There are high chances of communication failure between Portal and Database server. The probability of data unavailability is 40% as users are unable to view the tutorials. The cost of unavailability data will be measured in terms of cost of its communication establishment which is estimated as 8.

Countermeasure alternative1 and 2:

1) Replication of database to a parallel server in order to make the data available. This option will reduce the risk to 1% and is estimated to cost 3.

2) Poll the Database server every 1ms in order to maintain the communication This option will reduce the risk to 2% and estimated to cost 2.

$$RRL1 = (0.40 \times 8 - 0.01 \times 8) / 3 = (3.2 - 0.08) / 3 = 1.04$$

$$RRL2 = (0.40 \times 8 - 0.02 \times 8) / 2 = (3.2 - 0.16) / 2 = 1.52$$

Since $RRL2 > RRL1$ which indicates cost effective risk reduction measure. Second alternative is the better option.

Appendix

Task	Description	Time Taken(hours)
Task1	Identifying and finding inconsistencies in vision document	2
Task2	Documenting conflicts	3
Task3	Conflict resolution	3
Task4	Conflict evaluation	2
Task5	Risk management	4
	Total	14

